Quiz 6

1 Grading Scheme

- Question 1 is for 3 point
- Question 2 is for 2 points
- Bonus question 3 (No partial credit)

Total grade is 5 + Bonus.

2 Solutions

1.
   - $\binom{5}{0} = \frac{5!}{0!5!} = 1$
   - $\binom{5}{1} = \frac{5!}{1!4!} = 5$
   - $\binom{5}{2} = \frac{5!}{2!3!} = 5 \times 2 = 10$
2. We will show that if n is odd then, n^2 is odd. Suppose that n is odd, that is $n = 2a+1$ for some integer a. Then,
   $$n^2 = (2a+1)^2 = 4a^2 + 4a + 1 = 2(2a^2 + 2a) + 1$$
   Hence, n^2 is odd.
3. ($\Longrightarrow$) Suppose $0 \mid x$. Then there exist an integer c such that $c \times 0 = x$. Because $c \times 0 = 0$, we have $x = 0$.

   ($\Longleftarrow$) Suppose $x = 0$. As $c \times 0 = 0$ for every integer c, we have that $0 \mid x$.

3 Common error

For if-then proof, DO NOT write "we say A then B". Some of the natural ways to start an if then else proof are the following

1. Suppose A is true. We will show that B is true.
2. We will show that if A then B.
3. For the sake of contradiction, suppose not B. We will show not A.